

Homework 14

Please read Section 3.4 of Durrett.

1. Please work Exercise 28 from Chapter 3 of Durrett.

The number of children in a randomly chosen family has the following distribution:

# children	0	1	2	3	4
probability	0.15	0.25	0.3	0.2	0.1

Let A be the event that a family has exactly two girls. Durrett tells us to assume that if we are given the number of children a family has, we can treat the genders of the children (boy or girl, here) as independent with probability $1/2$ of boy and probability $1/2$ of girl.

We calculate $P(A)$ using the law of total probability. For $i = 0, 1, 2, 3, 4$, let B_i be the event that the family has i children. According to the table, the B_i form a partition. (Families only have 0, 1, 2, 3, or 4 children.)

$$P(A) = P(B_0)P(A|B_0) + P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3) + P(B_4)P(A|B_4)$$

The assumptions about the gender of children given the number of children means that assuming that B_i occurs, the number of girls in the family follows the binomial($i, 1/2$)

distribution. Therefore, $P(A|B_i) = 0$ for $i=0, 1$ and $P(A|B_i) = C_{i,2} \left(\frac{1}{2}\right)^i$ for $i=2, 3, 4$. We thus have

$$\begin{aligned} P(A) &= (0.15)(0) + (0.25)(0) + (0.3) \left(1 \cdot \left(\frac{1}{2}\right)^2\right) + (0.2) \left(3 \cdot \left(\frac{1}{2}\right)^3\right) + (0.1) \left(6 \cdot \left(\frac{1}{2}\right)^4\right) \\ &= \frac{0.3}{4} + \frac{0.6}{8} + \frac{0.6}{16} = \frac{3}{40} + \frac{6}{80} + \frac{6}{160} = \frac{12+12+6}{160} = \frac{30}{160} = \frac{3}{16}. \end{aligned}$$

2. Please work Exercise 34 from Chapter 3 of Durrett.

We use the Bayes'-formula method. Let A be the event that a randomly chosen person is color blind, and let B be the event that the person is a man.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

We calculate $P(A \cap B)$ using the multiplication rule (formula 3.3), and we calculate $P(A)$ using the law of total probability for the partition with B and B^c . (In Durrett's setting B^c is the event "the person is a woman.")

$$P(A \cap B) = P(B)P(A|B) = (0.5)(0.05) = 0.025$$

$$P(A) = P(B)P(A|B) + P(B^c)P(A|B^c) = (0.5)(0.05) + (0.5)(0.0025) = 0.025 + 0.00125$$

We thus have $P(B|A) = \frac{0.025}{0.025 + 0.00125} = \frac{20}{21} \approx 0.9524$.

3. Please work Exercise 45 from Chapter 3 of Durrett. The textbook answer is $4/5$. The following steps will help you explain this value:

a. Let A be the event that you get four heads in four flips. Let B be the event that you pick the trick coin, and let C be the event that you pick a regular coin. Durrett asks us to calculate $P(B|A)$, which is $P(A \cap B)/P(A)$. Follow the Bayes' formula method to write $P(A \cap B)$ and $P(A)$ in terms of $P(B)$, $P(C)$, $P(A|B)$, and $P(A|C)$:

$$P(A \cap B) = P(B) P(A|B)$$

$$P(A) = P(B) P(A|B) + P(C) P(A|C)$$

b. Calculate $P(A|B)$. To do this, use the "reducing the sample space" strategy. Assume you are flipping the trick coin. What is the probability of getting four heads?

$$P(A|B) = 1$$

c. Calculate $P(A|C)$. To do this, use the "reducing the sample space" strategy. Assume you are flipping a regular coin. What is the probability of getting four heads?

$$P(A|C) = 1/16$$

d. Combine your values for $P(A|B)$ and $P(A|C)$ with the values for $P(B)$ and $P(C)$ to find $P(B|A)$.

$$P(B|A) = \frac{\left(\frac{1}{5}\right)(1)}{\left(\frac{1}{5}\right)(1) + \left(\frac{4}{5}\right)\left(\frac{1}{16}\right)} = \frac{16}{16+4} = \frac{16}{20} = \frac{4}{5}.$$

4. Please work Exercise 54 from Durrett.

Durrett tells us that three bags are on table. One bag has two gold coins, one has two silver coins, and one has a silver coin and a gold coin. We pick a bag at random—equally likely to pick any of the three bags—and pick out a coin. The coin is gold. We calculate the probability that we picked from the bag with two gold coins.

Let A be the event that we picked a gold coin. Let B_1 be the event that we picked from the bag with two gold coins, B_2 be the event that we picked from the bag with two silver coins, and B_3 be the event that we picked from the bag with one gold and one silver coin. The events B_1, B_2, B_3 form a partition.

We use the Bayes' formula strategy to calculate $P(B_1|A) = \frac{P(B_1 \cap A)}{P(A)}$.

$$P(B_1 \cap A) = P(B_1)P(A|B_1) = \left(\frac{1}{3}\right)(1)$$

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3) = \left(\frac{1}{3}\right)(1) + \left(\frac{1}{3}\right)(0) + \left(\frac{1}{3}\right)\left(\frac{1}{2}\right)$$

We thus have

$$P(B_1|A) = \frac{\frac{1}{3}}{\frac{1}{3} + \left(\frac{1}{3}\right)\left(\frac{1}{2}\right)} = \frac{2}{2+1} = \frac{2}{3}.$$

Here is another way to think of this question: To calculate $P(B_1|A)$, we reduce the sample space by conditioning on A . There are three gold coins among the three bags. We are equally likely to have picked any of the three. Two of the coins came from bag 1 and one came from bag 3. Therefore, the probability that we picked a coin from bag 1 is $2/3$. Durrett could have given us the same question as a straight conditional probability calculation before we learned all these new methods.

5. Please work Exercise 60 from Durrett.

Y	X = 1	2	3	4
1	0	1/12	1/12	1/12
2	1/12	0	1/12	1/12
3	1/12	1/12	0	1/12
4	1/12	1/12	1/12	0